

ST O19: Statistical Methods for Bio-Computing – Internal Assignment

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Introduction

As of April 20, 2020, the 2019 novel coronavirus (renamed to COVID-19) has spread to more than 180 countries with more than 1,91,000 deaths and more than 2.7 million confirmed cases. COVID-19 has a death rate of 7.07% as of 20 April. Such outbreaks demand elucidation of taxonomic classification and origin of virus genomic sequence along with phylogenetic analysis. This is required for strategic planning, treatment, and further research.

Coronaviruses are single-stranded positive-sense-RNA viruses that are known to contain some of the largest virus genomes, up to around 32 kbp in length. Coronaviruses have high rates of homologous RNA recombination between the viral genomes. Furthermore, the large size of coronavirus genomes is thought to be able to accommodate mutations to genes. Within the COVID-19 strains, over 99% sequence similarity and a lack of diversity with these strains suggest a common lineage and source. After increases in the number of coronavirus genome sequences available following efforts to investigate the diversity, the family Coronaviridae now contains four genera. We have focused only on Betacoronavirus for this assignment. As said, Betacoronaviruses are one of four genera of coronaviruses of the subfamily Orthocoronavirinae in the family Coronaviridae, of the order Nidovirales.

* A significant part of this is taken from "Machine Learning using intrinsic genomic signatures for rapid classification of novel pathogens: COVID-19 case study", Gurjit S. Randhawa et al, bioRxiv, 2020, <https://doi.org/10.1101/2020.02.03.932350>

Objectives

- i) Construction of phylogenetic trees :
 - Using different distance metrics.
 - Using UPGMA and NJ algorithms
- ii) Obtain Return time distributions for monomers and dimers, for every sequence.
- iii) Estimate transition probability matrix and carry out simulations for every sequence.

Data Description

The entire dataset that is provided to us consists of fasta sequences of 24 genomes of nCOV downloaded from GISAID database. For the purpose of this assignment, we have selected the first 7 genomes.

The sequence data is categorized and stored by virus isolate ID (EPI_ISL_XXXXX). A unique virus isolate is defined by a combination of Virus Name, Passage History, Date of Harvest and Submitting Laboratory. Sequence data for each segment is assigned a unique accession number (EPIXXXX). The sequence names for the data used are given below:

Sequence Name	Number of terms	Label
BetaCoV/Chongqing/IVDC-C001/2020 EPI_ISL_408481	29884	1
BetaCoV/France/IDF0515/2020 EPI_ISL_408430	29863	2
BetaCov/France/IDF0626/2020 EPI_ISL_408431	29851	3
BetaCoV/Jiangxi/IVDC-JX-002/2020 EPI_ISL_408486	29876	4
BetaCoV/Shandong/IVDC-SD-001/2020 EPI_ISL_408482	29879	5
BetaCoV/Sichuan/IVDC-SC-001/2020 EPI_ISL_408484	29891	6
BetaCoV/Yunnan/IVDC-YN-003/2020 EPI_ISL_408480	29885	7

Question 1

- Aim:** i) To compute entropy for each sequence
 ii) To compute Mutual Information Content (MIC) between each pair of sequence

Entropy of a sequence

It is an uncertainty measure of a sequence. It is computed as

$$H(p_1, p_2, p_3, \dots, p_k) = -\sum_{i=1}^k p_i \log_2 p_i = E(-\log_2 p_i)$$

Where, p_i : Proportion corresponding to i^{th} class

For our sequences, $k = 4$.

For the computation purpose, R code is written and following are the results:

Seq No	P _A	P _C	P _G	P _T	Entropy
1	0.2990898	0.1837773	0.1961250	0.3210079	1.880036
2	0.2985969	0.1838060	0.1962629	0.3213341	1.880071
3	0.2985830	0.1839134	0.1962748	0.3212288	1.880293
4	0.2991364	0.1837930	0.1961441	0.3209265	1.880099
5	0.2991399	0.1837411	0.1960909	0.3210281	1.879948
6	0.2989863	0.1837342	0.1960456	0.1837342	1.879847
7	0.2990798	0.1837711	0.1961519	0.1837711	1.880045

Table 1

Comments:

Entropy of a sequence can be interpreted as the average level of information or uncertainty. Here, all the sequences have approximately equal entropies. Hence all the sequences have the same level of information averaged on all its possible outcomes.

Mutual Information Content

It is a measure of relationship between two sequences. For sequences X and Y, it is given by,

$$M(X, Y) = \sum_i \sum_j p_{ij} \log_2 \left(\frac{p_{ij}}{p_i p_j} \right)$$

Where,

$$p_{ij} = P[X = x_i, Y = y_j]$$

For finite sequences, it is a bounded function. MIC can be thought of as the reduction in uncertainty about one random variable given knowledge of another

For this metric, length of both the sequences need to be same. Hence, we have adjusted for the equal lengths to the minimum of the two sequence lengths.

i) For Complete Sequence

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]
[1,]	1.957124473	0.0017042128	0.0017154440	0.001073856	0.0021368792	0.0017596608	0.031793398
[2,]	0.001704213	1.9571944759	0.0013671025	0.001649924	0.0009778086	0.0316215375	0.002144292
[3,]	0.001715444	0.0013671025	1.9572879327	0.001714696	0.0317182694	0.0009622062	0.002136889
[4,]	0.001073856	0.0016499243	0.0017146958	1.957155409	0.0033877333	0.0024102748	0.003036788
[5,]	0.002136879	0.0009778086	0.0317182694	0.003387733	1.9570761971	0.0031732840	0.001715673
[6,]	0.001759661	0.0316215375	0.0009622062	0.002410275	0.0031732840	1.9569239557	0.001712589
[7,]	0.031793398	0.0021442920	0.0021368890	0.003036788	0.0017156733	0.0017125888	1.957137621

ii) For First 10% terms

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]
[1,]	1.978394496	0.001814097	0.002565685	0.002523603	0.004568966	0.006335077	0.033764464
[2,]	0.001814097	1.978160474	0.002724743	0.002360686	0.002131196	0.033926166	0.004512707
[3,]	0.002565685	0.002724743	1.978572781	0.002563920	0.034069981	0.002147049	0.004534569
[4,]	0.002523603	0.002360686	0.002563920	1.978705979	0.002508737	0.007140188	0.004018623
[5,]	0.004568966	0.002131196	0.034069981	0.002508737	1.978574070	0.005085131	0.001819305
[6,]	0.006335077	0.033926166	0.002147049	0.007140188	0.005085131	1.976015596	0.001759095
[7,]	0.033764464	0.004512707	0.004534569	0.004018623	0.001819305	0.001759095	1.978445810

iii) For Middle 10% terms

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]
[1,]	1.952292547	0.003524651	0.006013129	0.003577812	0.003898564	0.003436716	0.026525097
[2,]	0.003524651	1.952730700	0.004006014	0.002209890	0.003206033	0.003223232	0.004071704
[3,]	0.006013129	0.004006014	1.952418500	0.004111785	0.006093993	0.005359144	0.001876963
[4,]	0.003577812	0.002209890	0.004111785	1.951955788	0.026571326	0.003442731	0.004126350
[5,]	0.003898564	0.003206033	0.006093993	0.026571326	1.952362425	0.005284166	0.003558251
[6,]	0.003436716	0.003223232	0.005359144	0.003442731	0.005284166	1.952469355	0.003865669
[7,]	0.026525097	0.004071704	0.001876963	0.004126350	0.003558251	0.003865669	1.952469355

iv) For Last 10% terms

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]
[1,]	1.976351385	0.002672327	0.004187149	1.976351385	1.965461279	0.024481981	0.024327362
[2,]	0.002672327	1.978942706	0.023649869	0.002672327	0.002764207	0.005353549	0.005323539
[3,]	0.004187149	0.023649869	1.978981233	0.004187149	0.004093456	0.002782952	0.002699652
[4,]	1.976351385	0.002672327	0.004187149	1.976351385	1.965461279	0.024481981	0.024327362
[5,]	1.965461279	0.002764207	0.004093456	1.965461279	1.976351385	0.024624519	0.024469901
[6,]	0.024481981	0.005353549	0.002782952	0.024481981	0.024624519	1.976493021	1.972932472
[7,]	0.024327362	0.005323539	0.002699652	0.024327362	0.024469901	1.972932472	1.976540532

Comments:

The Mutual information content using the first or middle 10% of the terms is similar to the MIC calculated using the entire sequences. The last 10% show high MIC for certain pairs like (1,4), (1,5) and (6,7). This means that reduction in the uncertainty is large for the last 10% terms of 1st sequence if we have knowledge of last 10% terms of 4th or 5th sequence.

Question 2

Aim: To construct a phylogenetic tree using UPGMA

UPGMA

UPGMA (Unweighted Pair Group Method with Arithmetic Mean) is a simple method of tree construction, based on agglomerative hierarchical clustering algorithm. This method assumes the rate of substitution for all the branches in the tree is constant and produces an ultrametric tree, that is a tree having all the leaf nodes at equal length from the root. The most similar OTUs (simple or composite) are clustered together at each iteration.

1) Let x and y be two sequences. The distance between these two is calculated as

$$d(x, y) = \sqrt{(f_A^x - f_A^y)^2 + (f_G^x - f_G^y)^2 + (f_C^x - f_C^y)^2 + (f_T^x - f_T^y)^2}$$

where f_i^j = frequency of nucleotide i in sequence j ($i = A, G, T, C; j = x, y$)

This is an alignment free distance metric and is easy to compute. It relaxes the assumption of independent evolution of sites. The drawback of using this distance is that it is not context sensitive.

Iteration 1 :

Distance matrix calculated for the 7 sequences

	1	2	3	4	5	6	7
1	0.0000	21.4243	25.4755	5.2915	3.0000	9578.000	1.0000
2	21.4243	0.0000	8.3666	21.6564	21.4942	9581.021	21.4476
3	25.4755	8.3666	0.0000	24.0624	25.1794	9574.030	25.5734
4	5.2915	21.6564	24.0624	0.0000	4.3589	9573.000	5.5678
5	3.0000	21.4942	25.1794	4.3589	0.0000	9577.000	3.7417
6	9578.0001	9581.0214	9574.0303	9573.0001	9577.0003	0.000	9578.0003
7	1.0000	21.4476	25.5734	5.5678	3.7417	9578.000	0.0000

Minimum distance observed for pair (1,7), that is sequences 1 and 7 are clustered together. The distances are revised as follows :

$$d(17, k) = (d(1, k) + d(7, k))/2 \quad k = 2, 3 \dots 6$$

The branch length at this stage is $\frac{1}{2} = 0.5$. The phylogenetic tree at this stage (Representation, lengths not upto scale here).

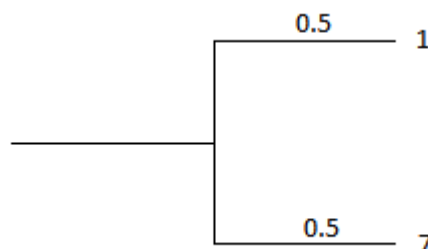


Figure 1

Iteration 2 :

Distance matrix calculated for 1 cluster and the remaining 6 sequences

	17	2	3	4	5	6
17	0.0000	21.4359	25.5245	5.4296	3.3708	9578.000
2	21.4359	0.0000	8.3666	21.6564	21.4942	9581.021
3	25.5245	8.3666	0.0000	24.0624	25.1794	9574.030
4	5.4296	21.6564	24.0624	0.0000	4.3589	9573.000
5	3.3708	21.4942	25.1794	4.3589	0.0000	9577.000
6	9578.0002	9581.0214	9574.0303	9573.0001	9577.0003	0.000

Minimum distance observed for pair of cluster(1,7) and sequence 5, thus is sequence 5 is clustered with the previous cluster. The distances are revised as follows :

$$d(175, k) = (d(17, k) + d(5, k))/2 \quad k = 2, 3, 4, 6$$

The branch length at this stage is $3.3708/2 = 1.6854$. The phylogenetic tree at this stage (Representation, lengths not upto scale here).

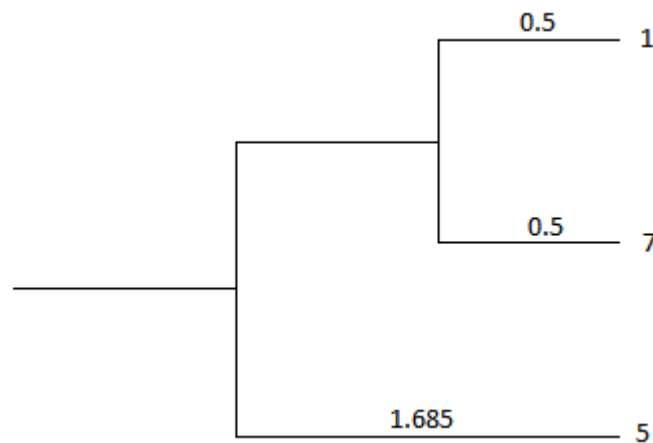


Figure 2

The above procedure is repeated until all the sequences are clustered together.

A representation of the tree (lengths not upto scale here) can be observed to be as follows :

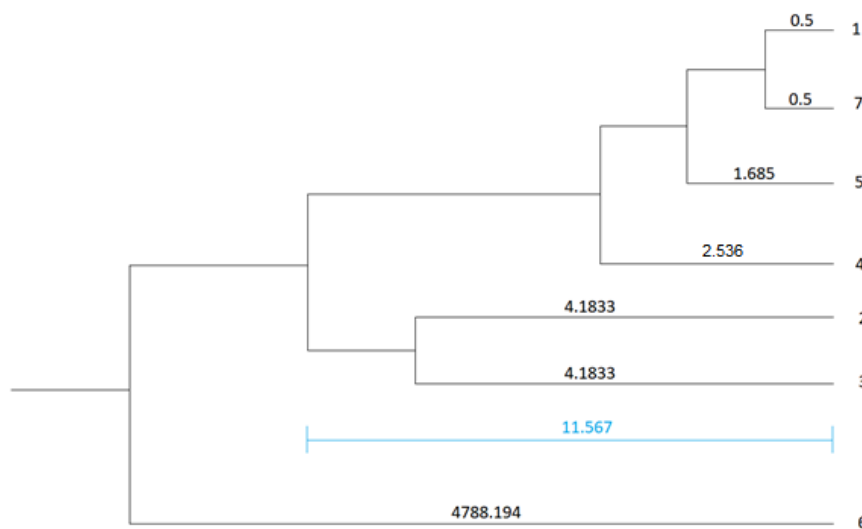


Figure 3

2) Let x and y be the two sequences. The distance between these two is calculated as

$$d(x, y) = |H(x) - H(y)|$$

where H : Entropy of the sequence = $\sum_{i \in S} -p_i \log_2 p_i$; $S = \{A, G, T, C\}$

This is an alignment free distance metric and is easy to compute.

Distance matrix calculated for 7 sequences. The distances are multiplied by the factor 10^5 for convenience.

	1	2	3	4	5	6	7
1	0.0000	0.7000	1.6346	0.3094	0.4828	2.0052	0.1315
2	0.7000	0.0000	0.9346	0.3907	1.1828	2.7052	0.5685
3	1.6346	0.9346	0.0000	1.3252	2.1174	3.6398	1.5031
4	0.3094	0.3907	1.3252	0.0000	0.7921	2.3145	0.1779
5	0.4828	1.1828	2.1174	0.7921	0.0000	1.5224	0.6142
6	2.0052	2.7052	3.6398	2.3145	1.5224	0.0000	2.1367
7	0.1315	0.5685	1.5031	0.1779	0.6142	2.1367	0.0000

The clustering of sequences is carried out using the “upgma()” function in R and the phylogenetic tree is plotted.

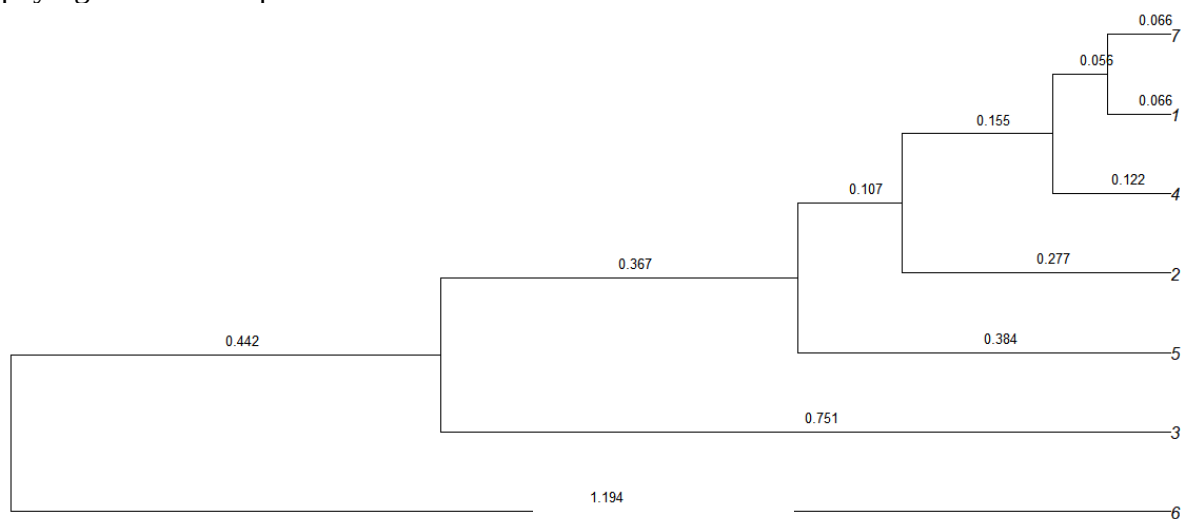


Figure 6

Comments :

The tree constructed using the given distance metric shows sequence 1 and 7 are the most similar and are hence clustered together. The composite cluster formed gets one sequence added to it every iteration.

3) Let x and y be the two sequences. The distance between these two is calculated using Juke – Cantor distance function, which is :

$$d(x, y) = \frac{3}{4} \log\left(1 - \frac{4}{3} f\right)$$

where f = proportion of u such that $x_u \neq y_u$

This is an alignment-based distance metric. Here the sequences are made to be of equal lengths by taking the length equal to the minimum of the two. This distance is less sensitive to the changes in the data as the log function moves slowly. Though this distance has high complexity in terms of memory, this does not pose a serious issue as here, the number of sequences being small.

Distance matrix calculated for 7 sequences.

	1	2	3	4	5	6	7
1	0.0000	2.4214	4.4019	3.5620	4.4390	4.2952	2.7267
2	2.4214	0.0000	3.6462	3.4202	3.2621	2.7543	4.5748
3	4.4019	3.6462	0.0000	4.3785	2.7435	3.2775	4.5895
4	3.5620	3.4202	4.3785	0.0000	2.2757	2.4186	2.2266
5	4.4390	3.2621	2.7435	2.2757	0.0000	2.3119	2.4126
6	4.2952	2.7543	3.2775	2.4186	2.3119	0.0000	2.4140
7	2.7267	4.5748	4.5895	2.2266	2.4126	2.4140	0.0000

The clustering of sequences is carried out using the “upgma()” function in R and the phylogenetic tree is plotted.

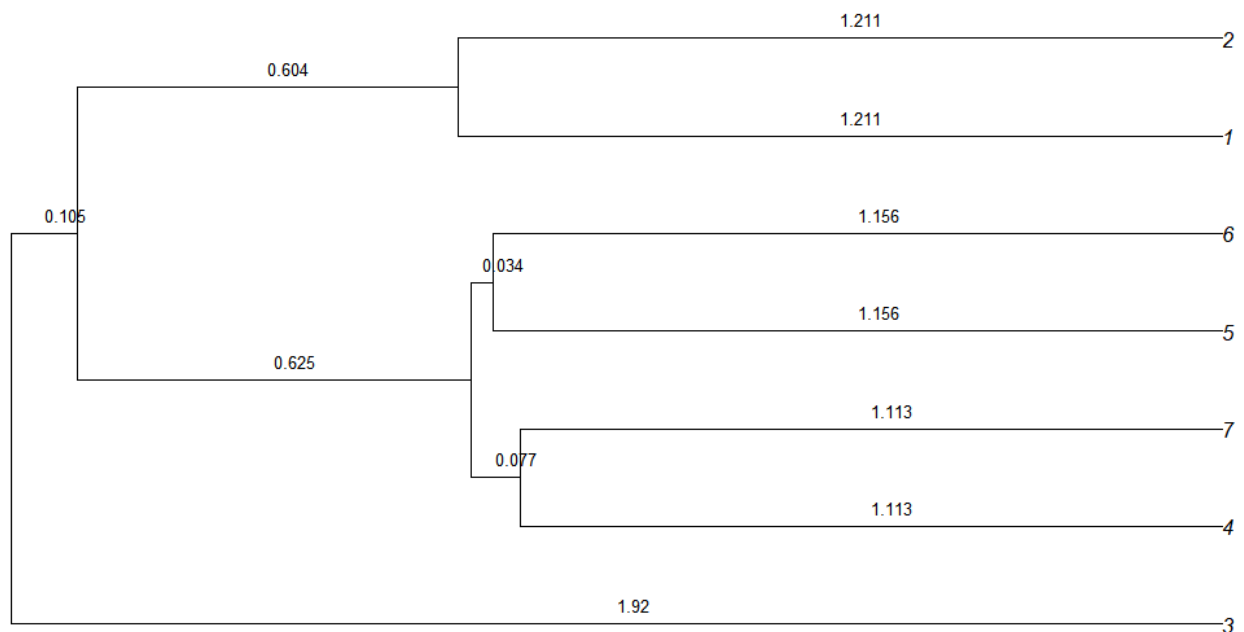


Figure 7

Comments :

The tree constructed using the given distance metric shows sequence 4 and 7 are the most similar and are hence clustered together.

Question 3

Aim: To use Mutual Information Content for obtaining Tree Topology.

Mutual information describes the amount of information shared by stochastic processes. It can thus be used to derive distance measures quantifying the similarity of the processes.

[Dawy et al. \[2005\]](#) have proposed a universal well-founded distance measure based on MIC. The said measure is also shown to be a metric.

Let X and Y be two sequences under consideration. Then,

$$d_{ij} = 1 - \frac{M(X,Y)}{\max(H(X), H(Y))}$$

Where, M is MIC and H is an entropy of a sequence.

Distance matrix calculated using above formula is given below.

	[,1]	[,2]	[,3]	[,4]	[,5]	[,6]	[,7]
[1,]	0.0000000	0.9990935	0.9990877	0.9994288	0.9988634	0.9990640	0.9830890
[2,]	0.9990935	0.0000000	0.9992729	0.9991224	0.9994799	0.9831807	0.9988595
[3,]	0.9990877	0.9992729	0.0000000	0.9990881	0.9831312	0.9994883	0.9988635
[4,]	0.9994288	0.9991224	0.9990881	0.0000000	0.9981981	0.9987180	0.9983848
[5,]	0.9988634	0.9994799	0.9831312	0.9981981	0.0000000	0.9983120	0.9990874
[6,]	0.9990640	0.9831807	0.9994883	0.9987180	0.9983120	0.0000000	0.9990891
[7,]	0.9830890	0.9988595	0.9988635	0.9983848	0.9990874	0.9990891	0.0000000

A phylogenetic tree is obtained using NJ method.

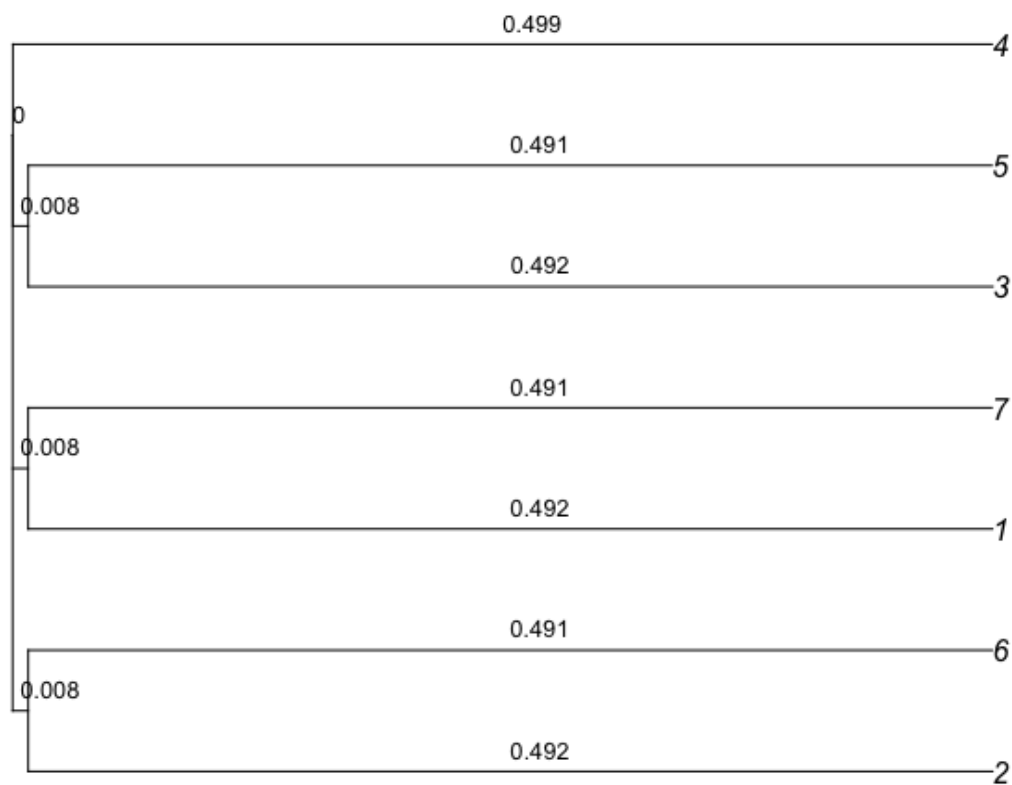


Figure 8

Comments:

The NJ method using the given distance metric gives an unrooted phylogenetic tree. All the branches are observed to have approximately the same length. This means that all sequences show equal level of similarity.

Question 4

- Aim :** i) To verify whether the selected distance is ultrametric.
ii) To obtain tree topology using Neighbour Joining method

Ultrametric condition:

A distance function satisfies the ultrametric condition if for every triplet of distances : d_{ij}, d_{jk} and d_{ik} , either all three distances are equal, or two are equal and the remaining is smaller. This means for an ultrametric distance the distance from the root to the leaf node (tips) is same for all leaf nodes.

NJ Method :

The NJ (Neighbour Joining) method is a method for reconstructing phylogenetic trees. The principle of this method is to find pairs of OTUs that minimise the total branch length at each stage of clustering, starting with a starlike tree. This method does not assume the rate of substitution to be constant and does not require the data to be ultrametric, unlike UPGMA method.

- 1) Let x and y be the two sequences. The distance between these two is calculated as

$$d(x, y) = \sqrt{(f_A^x - f_A^y)^2 + (f_G^x - f_G^y)^2 + (f_C^x - f_C^y)^2 + (f_T^x - f_T^y)^2}$$

where f_i^j = frequency of nucleotide i in sequence j ($i = A, G, T, C; j = x, y$)

The advantages and disadvantages of using this distance mentioned in Q3.

Distance matrix calculated for 7 sequences.

	1	2	3	4	5	6	7
1	0.0000	21.4243	25.4755	5.2915	3.0000	9578.000	1.0000
2	21.4243	0.0000	8.3666	21.6564	21.4942	9581.021	21.4476
3	25.4755	8.3666	0.0000	24.0624	25.1794	9574.030	25.5734
4	5.2915	21.6564	24.0624	0.0000	4.3589	9573.000	5.5678
5	3.0000	21.4942	25.1794	4.3589	0.0000	9577.000	3.7417
6	9578.0001	9581.0214	9574.0303	9573.0001	9577.0003	0.000	9578.0003
7	1.0000	21.4476	25.5734	5.5678	3.7417	9578.000	0.0000

The above distance matrix shows that the ultrametric condition is not satisfied by this distance function. Consider the triplet $d_{12} = 21.424$, $d_{23} = 8.367$ and $d_{13} = 25.476$. This triplet does not satisfy the required condition of having two distances to be equal.

The process of Neighbour Joining (NJ) method starts with a phylogenetic tree in the form of a star, that is all the sequences at equal distances and with a common node.

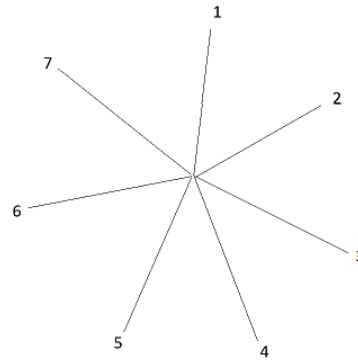


Figure 9

Iteration 1 :

$N = 7$.

The vector r is calculated as follows.

$$r(k) = \frac{\sum_{i \in S} d(k, i)}{N - 2}$$

where $S = \{1, 2 \dots 7\}$ and $N = \text{No. of sequences or clusters considered ; } k = 1, 2 \dots 7$

For the first iteration, $r = [1926.838 \ 1935.082 \ 1936.538 \ 1926.787 \ 1926.955 \ 11492.210 \ 1927.066]$

The distance matrix D is modified as follows : (MD : Modified Distance matrix)

$$MD(i, j) = D(i, j) - (r(i) + r(j)) \quad \text{where } i, j = 1, 2 \dots 7$$

The modified distance matrix for this iteration is :

	1	2	3	4	5	6	7
1	0.000	-3840.496	-3837.900	-3848.334	-3850.793	-3841.049	-3852.904
2	-3840.496	0.000	-3863.253	-3840.213	-3840.543	-3846.271	-3840.701
3	-3837.900	-3863.253	0.000	-3839.263	-3838.313	-3854.718	-3838.030
4	-3848.334	-3840.213	-3839.263	0.000	-3849.383	-3845.998	-3848.286
5	-3850.793	-3840.543	-3838.313	-3849.383	0.000	-3842.165	-3850.279
6	-3841.049	-3846.271	-3854.718	-3845.998	-3842.165	0.000	-3841.276
7	-3852.904	-3840.701	-3838.030	-3848.286	-3850.279	-3841.276	0.000

The pair (2,3) has the minimum modified distance. Hence sequence 2 and 3 form the first cluster.

$$U_1 = (2, 3)$$

The branch lengths for U_1 node are calculated as follows :

$$d(U_1, 2) = \frac{D(2, 3) + (r(2) - r(3))}{2} \quad \text{and} \quad d(U_1, 3) = \frac{D(2, 3) + (r(3) - r(2))}{2}$$

Here, $d(U_1, 2) = 3.455587$ and $d(U_1, 3) = 4.911014$.

The updated tree is given below. (Representation, lengths not upto scale here)

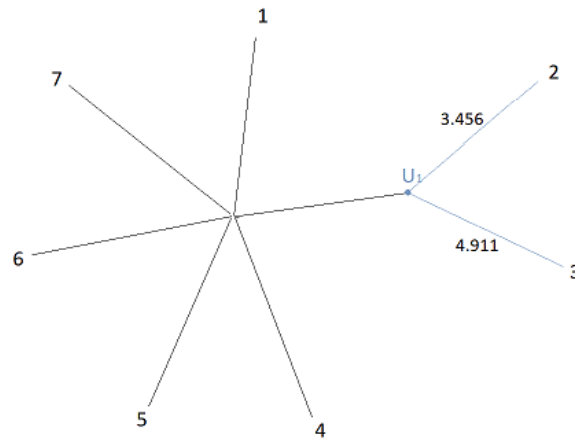


Figure 10

The distance matrix is updated by considering the distances of the remaining sequences to the node U_1 as

$$D(U_1, k) = D(2, k) + D(3, k) - D(2, 3) \text{ where } k = 1, 4, 5, 6, 7$$

Iteration 2 :

The new distance matrix is :

	1	$U_1 = 23$	4	5	6	7
1	0.0000	19.2666	5.2915	3.0000	9578.000	1.0000
$U_1 = 23$	19.2666	0.0000	18.6761	19.1535	9573.343	19.3272
4	5.2915	18.6761	0.0000	4.3589	9573.000	5.5678
5	3.0000	19.1535	4.3589	0.0000	9577.000	3.7417
6	9578.0001	9573.3426	9573.0001	9577.0003	0.000	9578.0003
7	1.0000	19.3272	5.5678	3.7417	9578.000	0.0000

R vector : $r = [2401.640 \ 2412.441 \ 2401.724 \ 2401.814 \ 11969.836 \ 2401.909]$

The Modified distance matrix, MD :

	1	23	4	5	6	7
1	0.000	-4794.814	-4798.072	-4800.453	-4793.475	-4802.549
23	-4794.814	0.000	-4795.489	-4795.102	-4808.935	-4795.023
4	-4798.072	-4795.489	0.000	-4799.178	-4798.559	-4798.065
5	-4800.453	-4795.102	-4799.178	0.000	-4794.649	-4799.981
6	-4793.475	-4808.935	-4798.559	-4794.649	0.000	-4793.745
7	-4802.549	-4795.023	-4798.065	-4799.981	-4793.745	0.000

The pair $(U_1, 6)$ has the minimum modified distance. Hence sequence 6 and node U_1 form the second cluster.

$$U_2 = (U_1, 6)$$

Here, $d(U_2, U_1) = 7.974116$ and $d(U_2, 6) = 9565.368$

The updated tree is given below. (Representation, lengths not upto scale here)

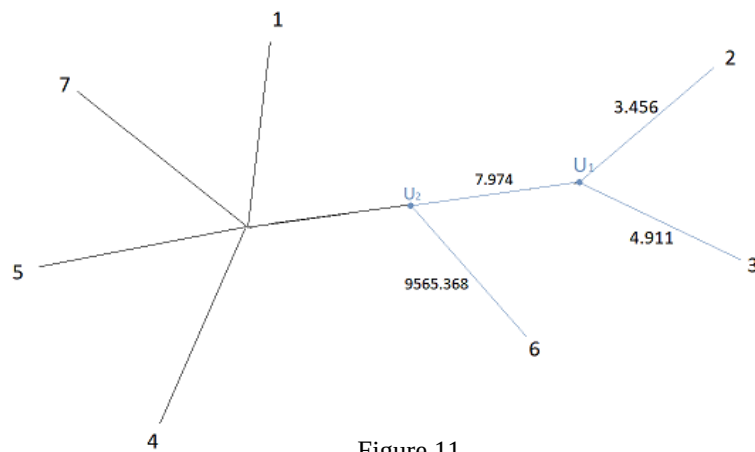


Figure 11

The above procedure is repeated until all the sequences are clustered together.

A representation of the tree (lengths not upto scale here) can be observed to be as follows :

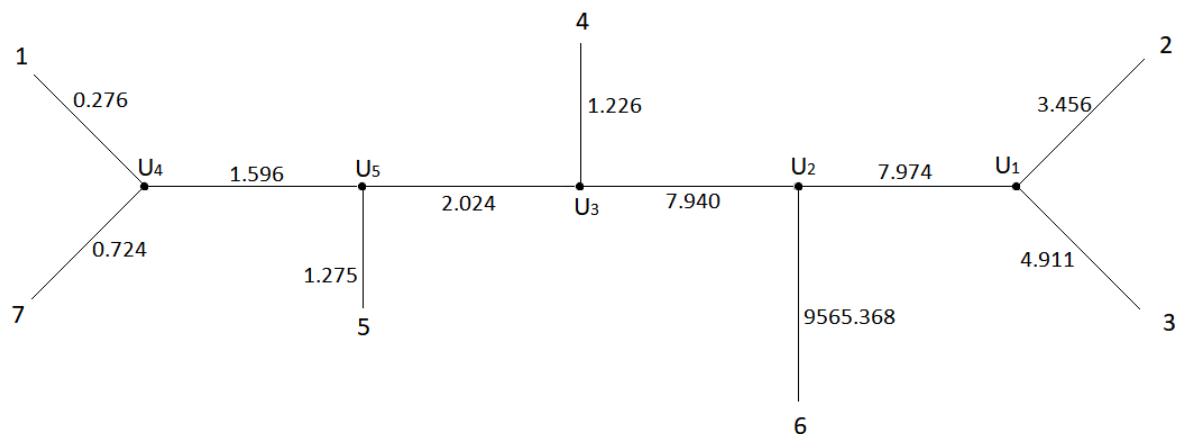


Figure 12

The actual tree is constructed below (Using R package “phangorn”, function “nj()”), where the lengths are according to the scale. (Connector B added for convenience of visualisation.)

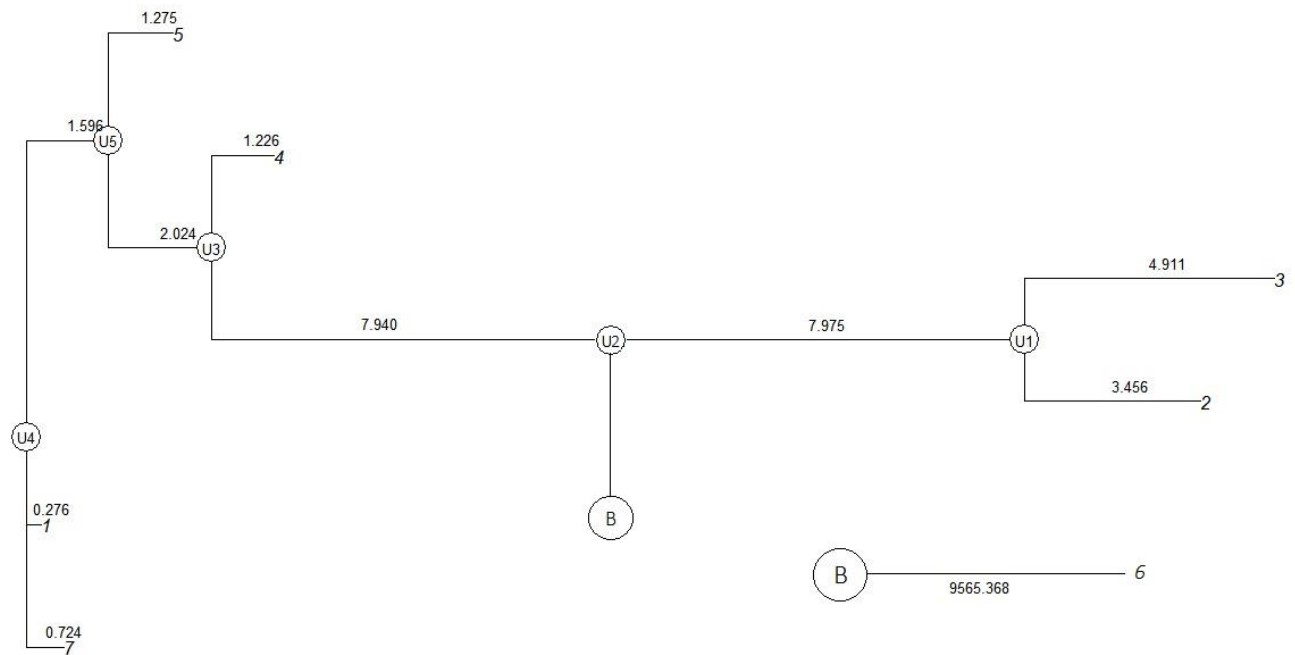


Figure 13

Comments :

The NJ method using the given distance metric gives an unrooted phylogenetic tree. The sequences 1 and 7 are observed to be the most similar and sequence 6 is the one that is the most dissimilar, which is evident from the very long branch length.

2) Let x and y be the two sequences.

The distance between the two is calculated using Juke - Cantor distance function, which is :

$$d(x, y) = \frac{3}{4} \log\left(1 - \frac{4}{3} f\right)$$

where f = proportion of u such that $x_u \neq y_u$

The advantages and disadvantages of using this distance mentioned in Q3.

Distance matrix calculated for 7 sequences.

	1	2	3	4	5	6	7
1	0.0000	2.4214	4.4019	3.5620	4.4390	4.2952	2.7267
2	2.4214	0.0000	3.6462	3.4202	3.2621	2.7543	4.5748
3	4.4019	3.6462	0.0000	4.3785	2.7435	3.2775	4.5895
4	3.5620	3.4202	4.3785	0.0000	2.2757	2.4186	2.2266
5	4.4390	3.2621	2.7435	2.2757	0.0000	2.3119	2.4126
6	4.2952	2.7543	3.2775	2.4186	2.3119	0.0000	2.4140
7	2.7267	4.5748	4.5895	2.2266	2.4126	2.4140	0.0000

The above distance matrix shows that the ultrametric condition is not satisfied by this distance function. Consider the triplet $d_{12} = 2.421$, $d_{23} = 3.646$ and $d_{13} = 4.402$. This triplet does not satisfy the required condition of having two distances to be equal.

The clustering of sequences is carried out using the “nj()” function in R and the phylogenetic tree is plotted.

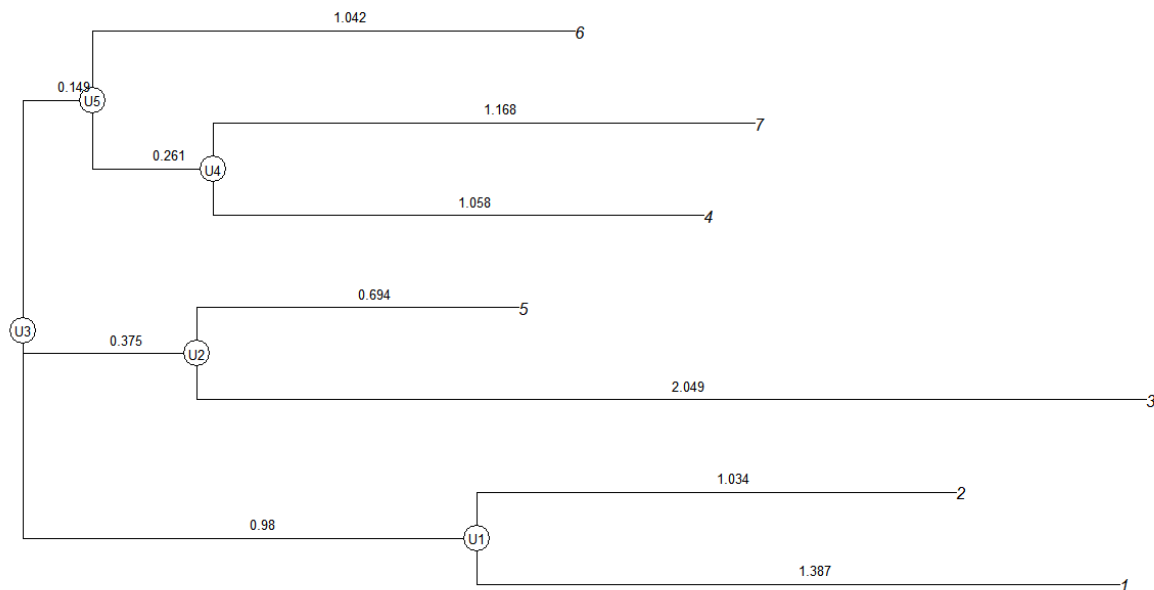


Figure 14

Comments :

The NJ method using the given distance metric gives an unrooted phylogenetic tree. The sequences 4 and 7 are observed to be the most similar here.

3) Let x and y be the two sequences.

The distance between these two is calculated using the distance function:

$$d(x, y) = \# u \text{ such that } x_u \neq y_u$$

This distance is easy to calculate and understand. It takes context sensitivity into account. It works on the assumption of independent evolution of sites. This is an alignment-based distance metric, hence the sequences are made to be of equal lengths by taking the length equal to the minimum of the two.

Distance matrix calculated for 7 sequences.

	1	2	3	4	5	6	7
1	0	21510	22325	22213	22349	22340	21822
2	21510	0	22215	22163	22108	21828	22347
3	22325	22215	0	22323	21811	22105	22339
4	22213	22163	22323	0	21329	21516	21256
5	22349	22108	21811	21329	0	21382	21511
6	22340	21828	22105	21516	21382	0	21517
7	21822	22347	22339	21256	21511	21517	0

The above distance matrix shows that the ultrametric condition is not satisfied by this distance function. Consider the triplet $d_{17} = 21822$, $d_{16} = 22340$ and $d_{67} = 21517$. This triplet does not satisfy the required condition of having two distances to be equal, and remaining smaller.

The clustering of sequences is carried out using the “nj()” function in R and the phylogenetic tree is plotted.

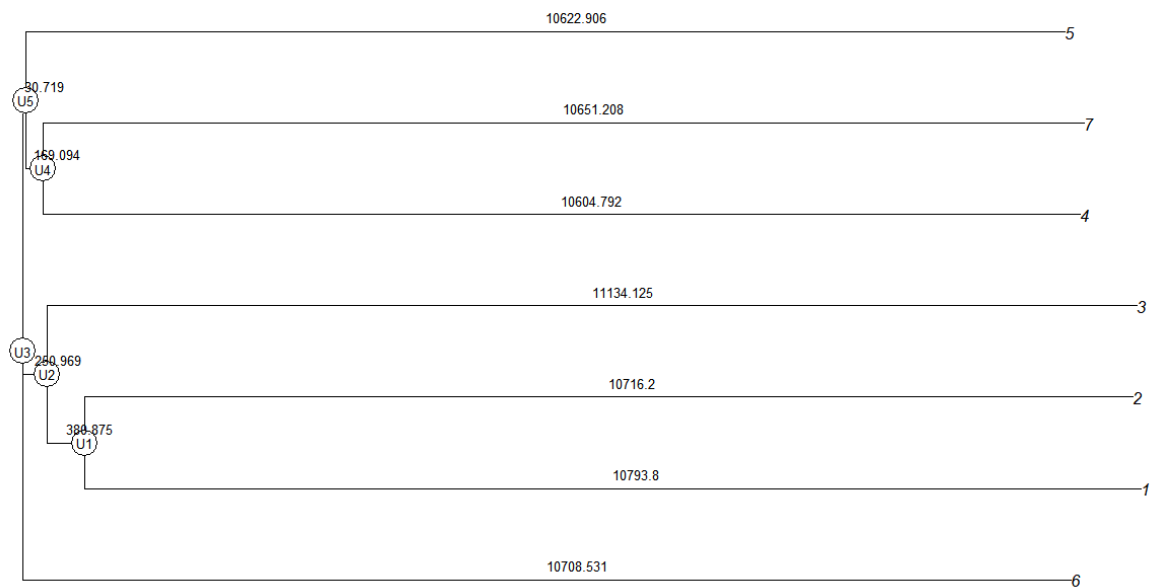


Figure 15

Comments :

The NJ method using the given distance metric gives an unrooted phylogenetic tree. All the sequences are observed to be clustered with large branch lengths, thus it can be said that no pair shows high level of similarity.

Question 5

- Aim:** i) To obtain return time frequency distribution for each sequence (mono and di levels)
ii) To use this distribution for tree topology

Return Time Frequency Distribution:

Return time for k-mer is defined as number of steps until the occurrence of next.
For sequence AGATTAGCCTATA, return times for 'A' are 1,3,4 and 1.

For Mono level:

Given a sequence, the return time distribution for the monomers is found as follows:

Step 1 :

Identify the sites of the sequence with a particular nucleotide S_1 . Calculate the step sizes (difference between the consecutive sites - 1) in this sequence, that is the number of sites required each time to get S_1 again.

Step 2 :

Find the distribution that fits the best to the sequence obtained above.

Step 3 :

Repeat for all 4 nucleotides to get the return time distributions for all monomers.

Step 4 :

Repeat the entire process for all the sequences.

Sequence 1:

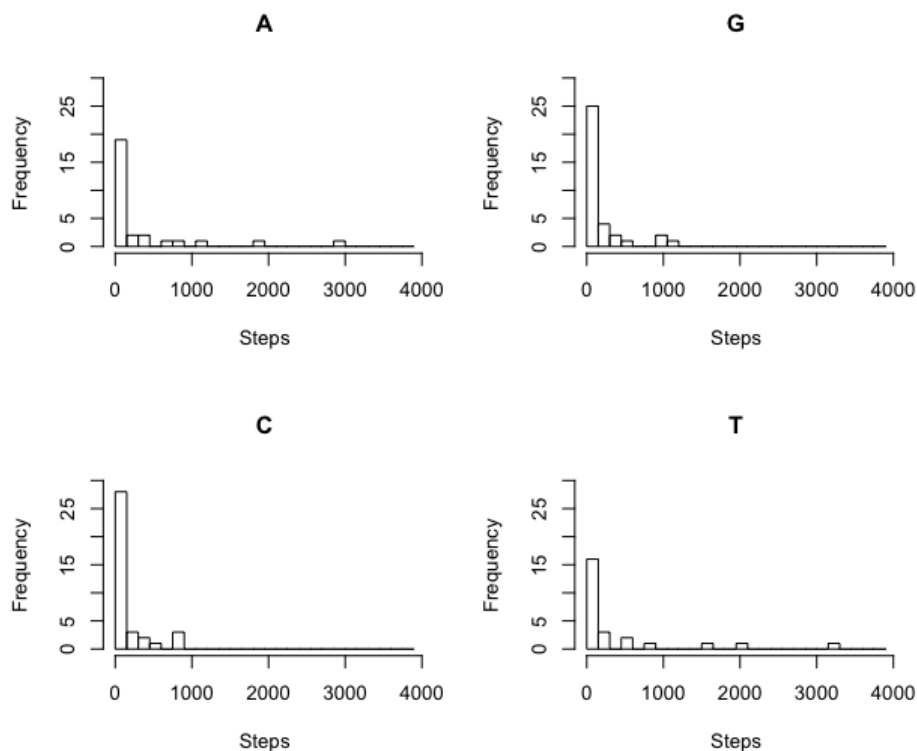


Figure 16

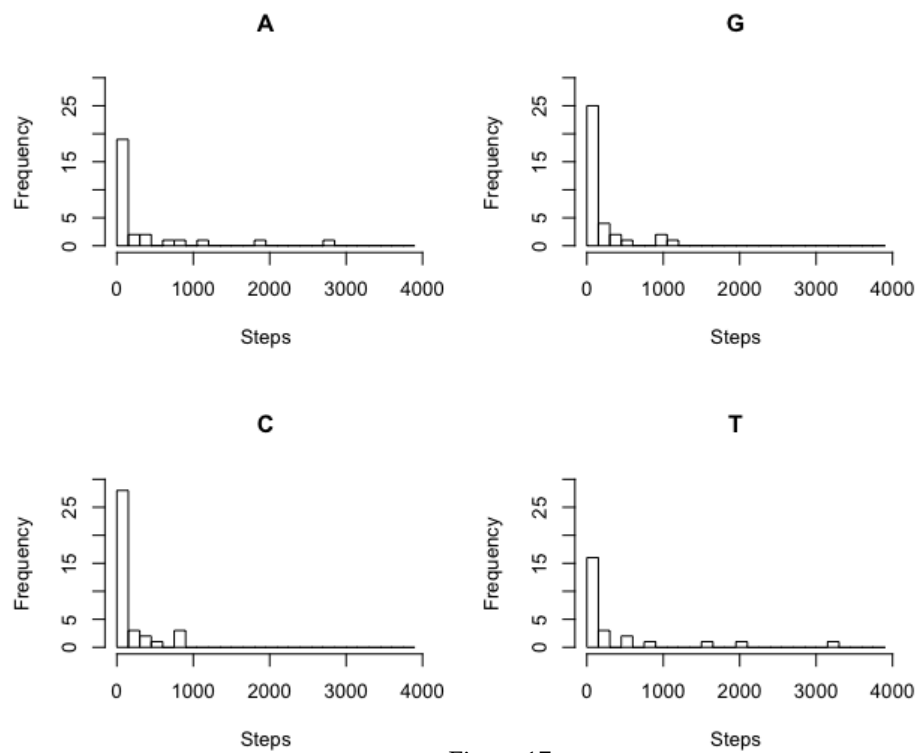
Sequence 2:

Figure 17

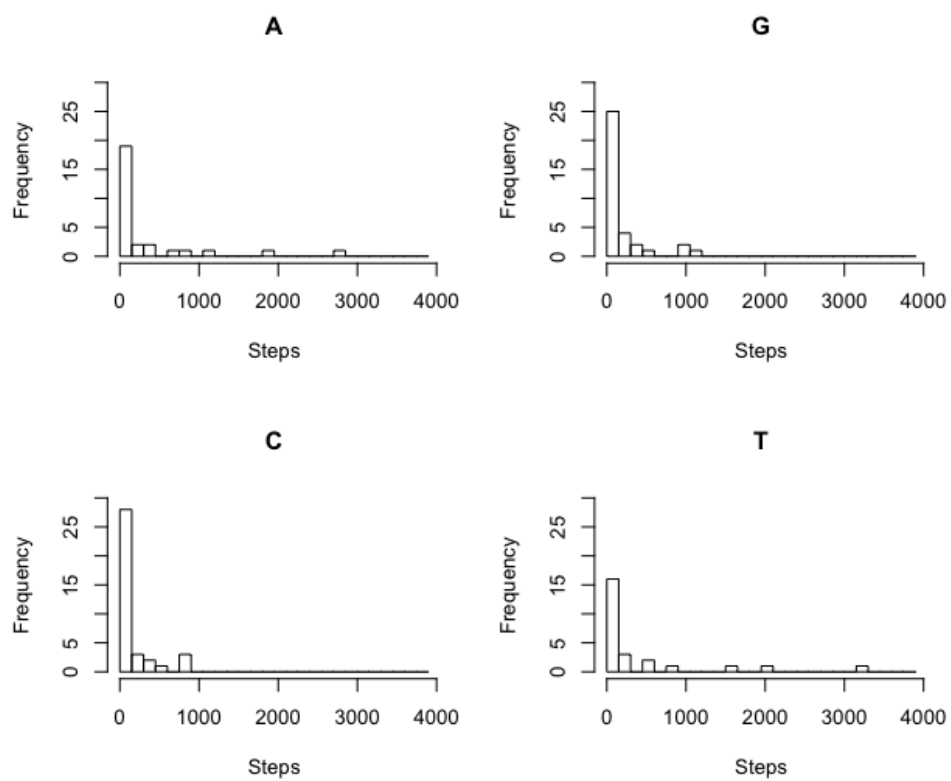
Sequence 3:

Figure 18

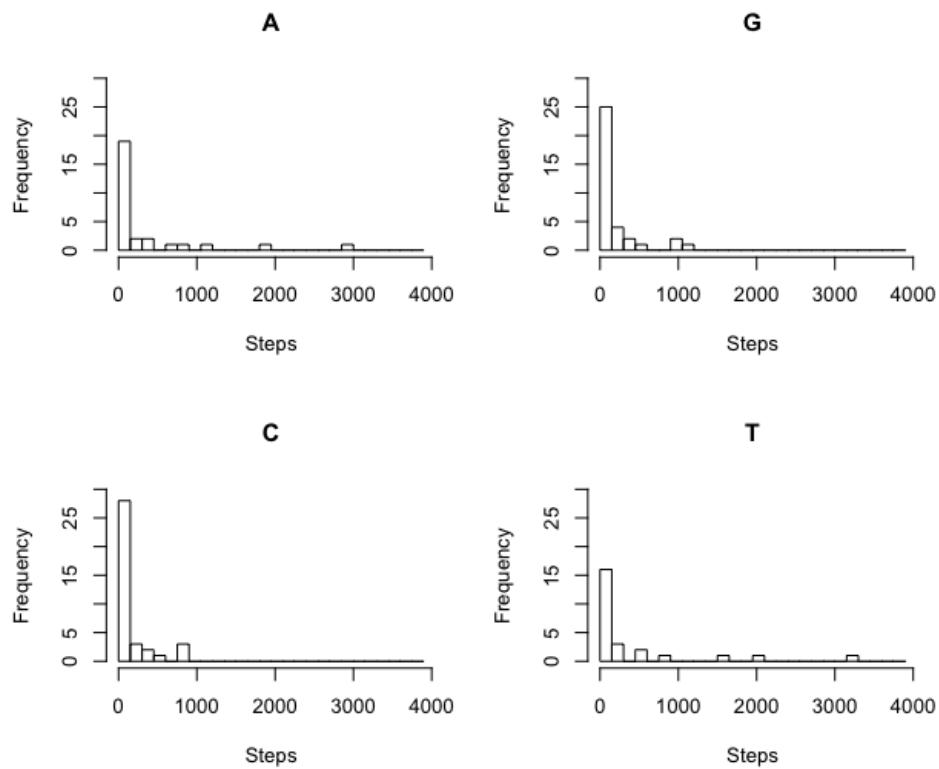
Sequence 4:

Figure 19

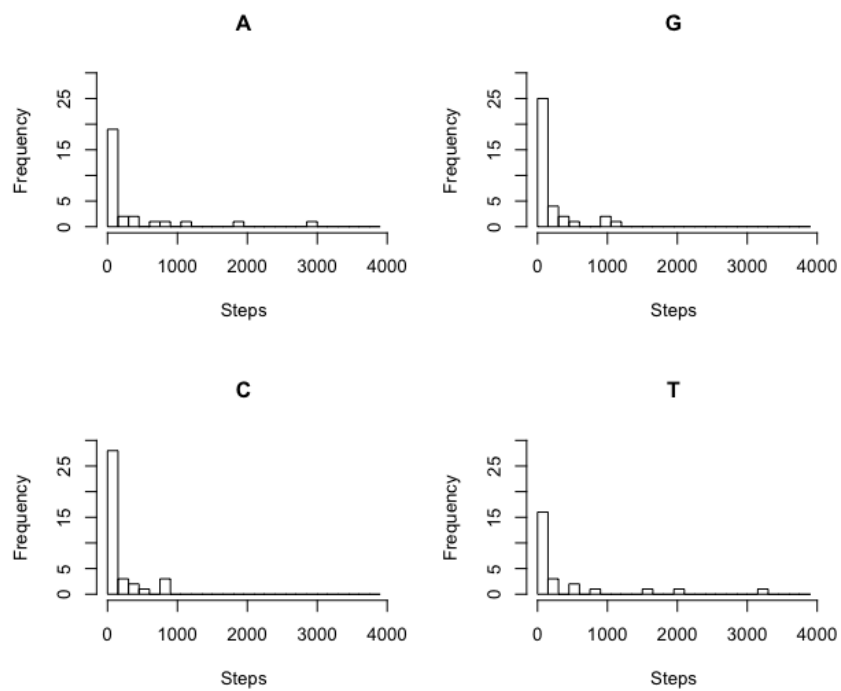
Sequence 5:

Figure 20

Figure 21

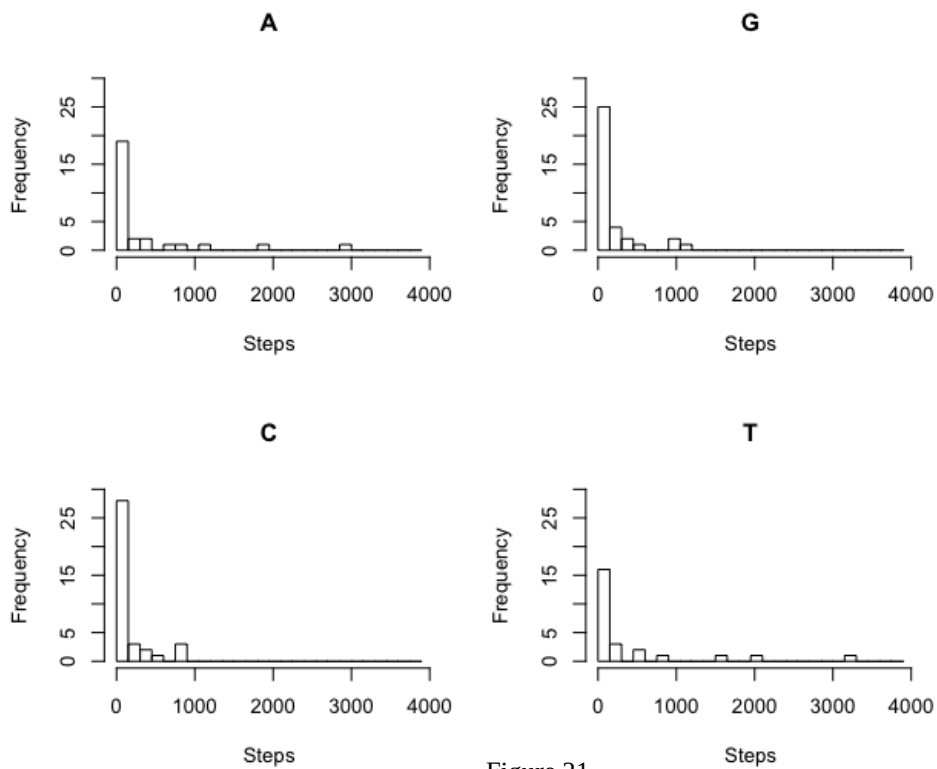
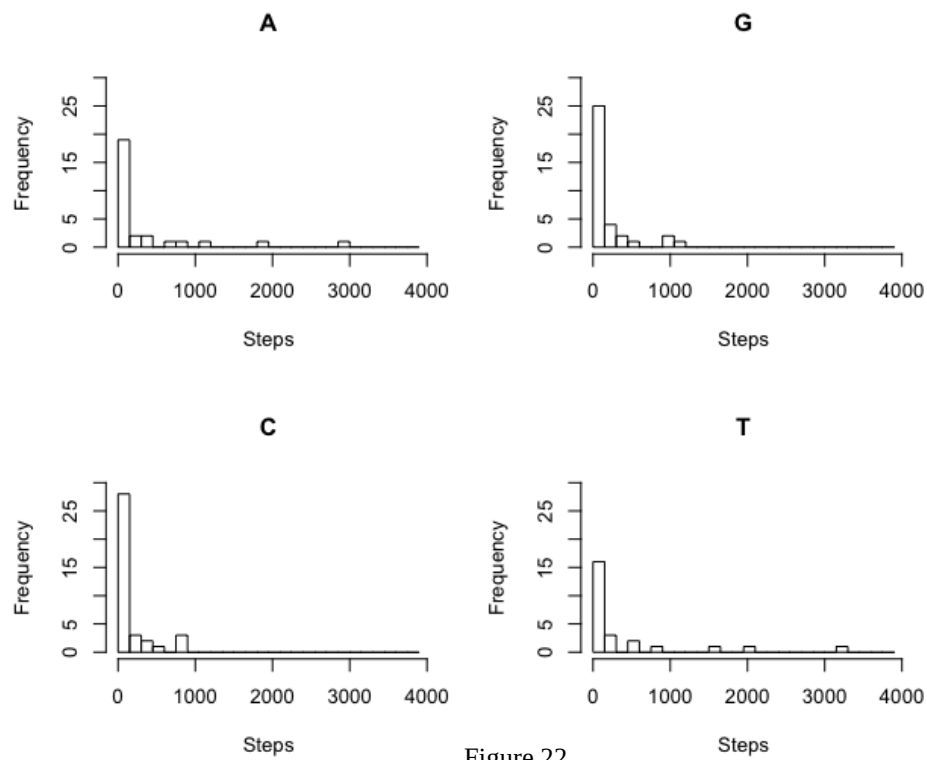


Figure 22



In all the cases, geometric distribution turned out to be a good fit. As an illustration, we have given return time distribution of ‘T’ for sequence 1.

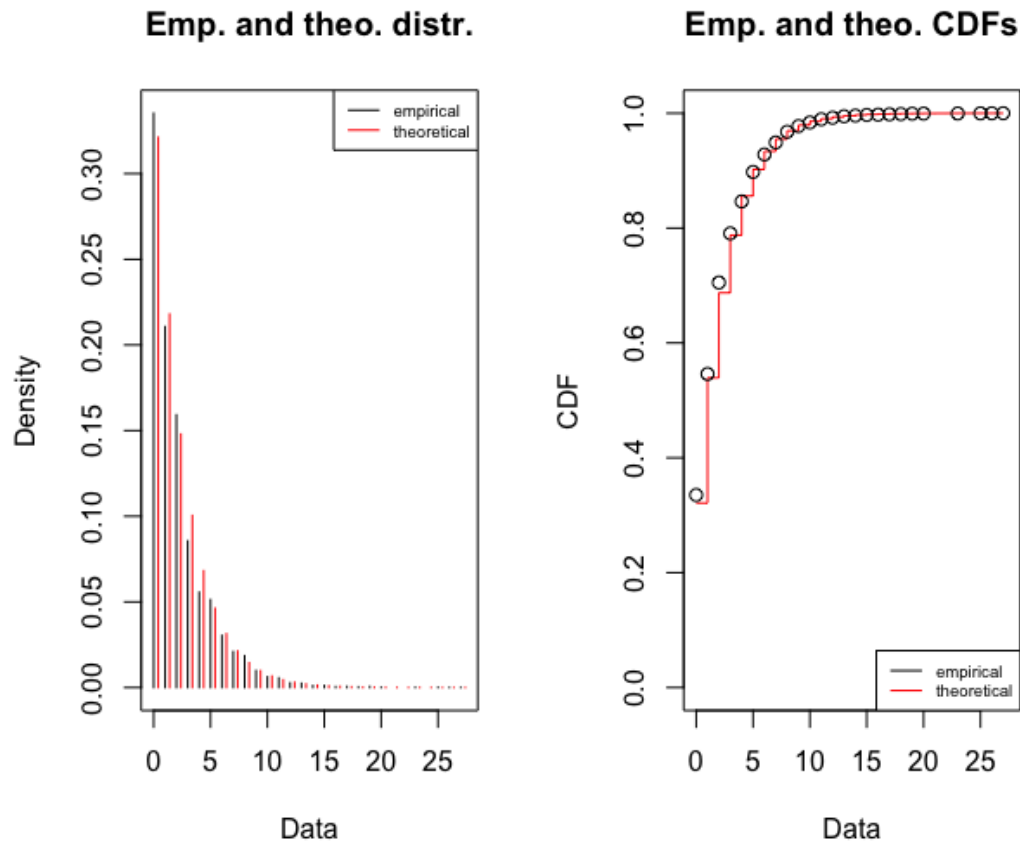


Figure 23

Comments:

The return time distributions of all the nucleotides (monomers) for all the sequences is observed to be same. There is no significant difference between all the sequences, which is evident from the entropy.

For Di level:

The return time distributions for dimers are obtained by a process similar to that for monomers.

As an illustration, return time distributions of dimers for sequence 1 are plotted.

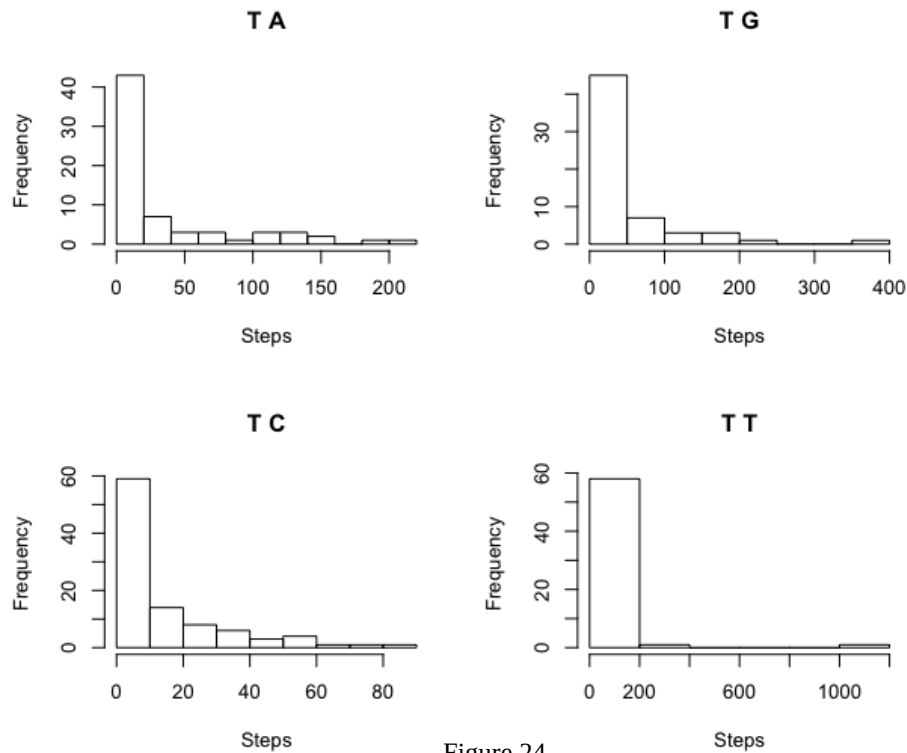
Sequence 1:

Figure 24

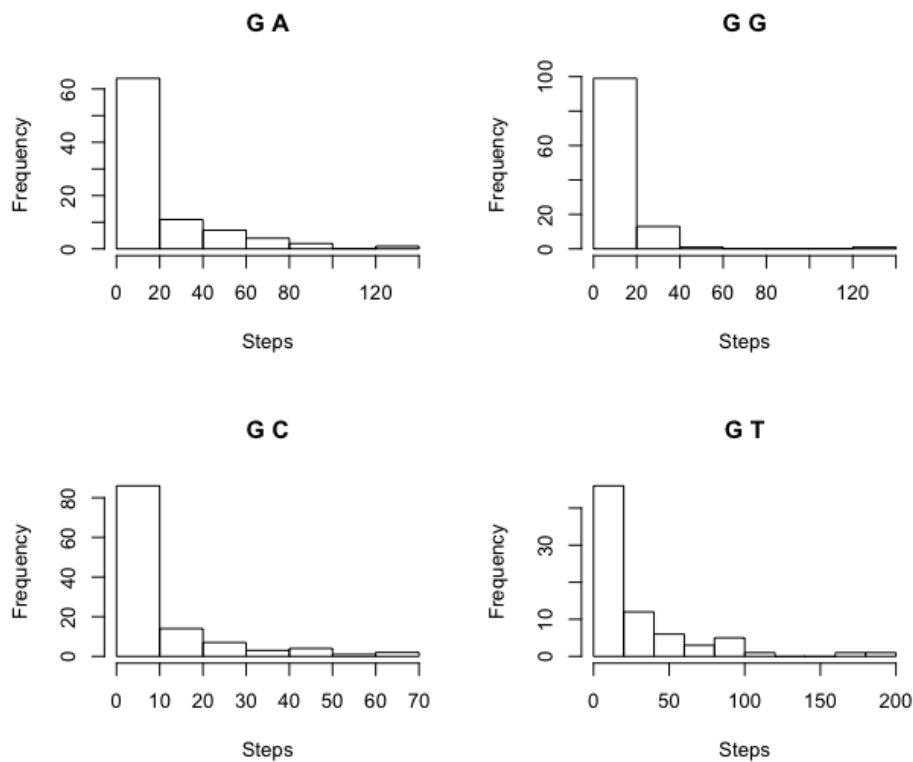


Figure 25

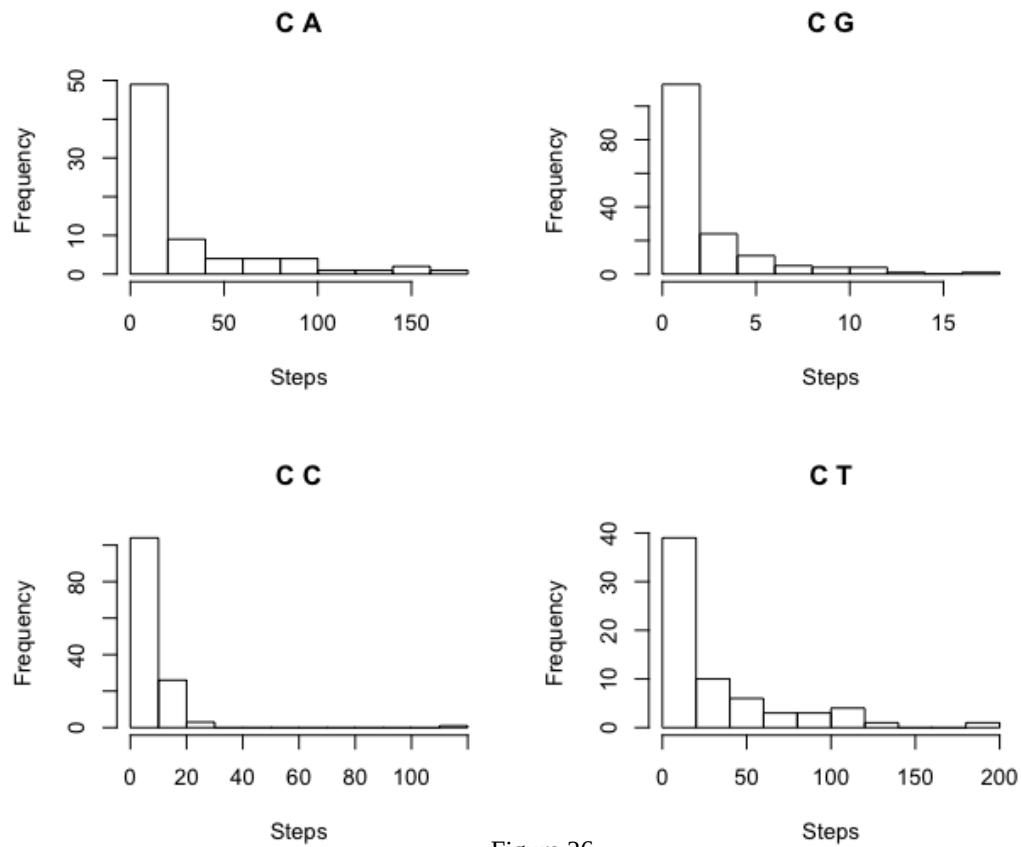


Figure 26

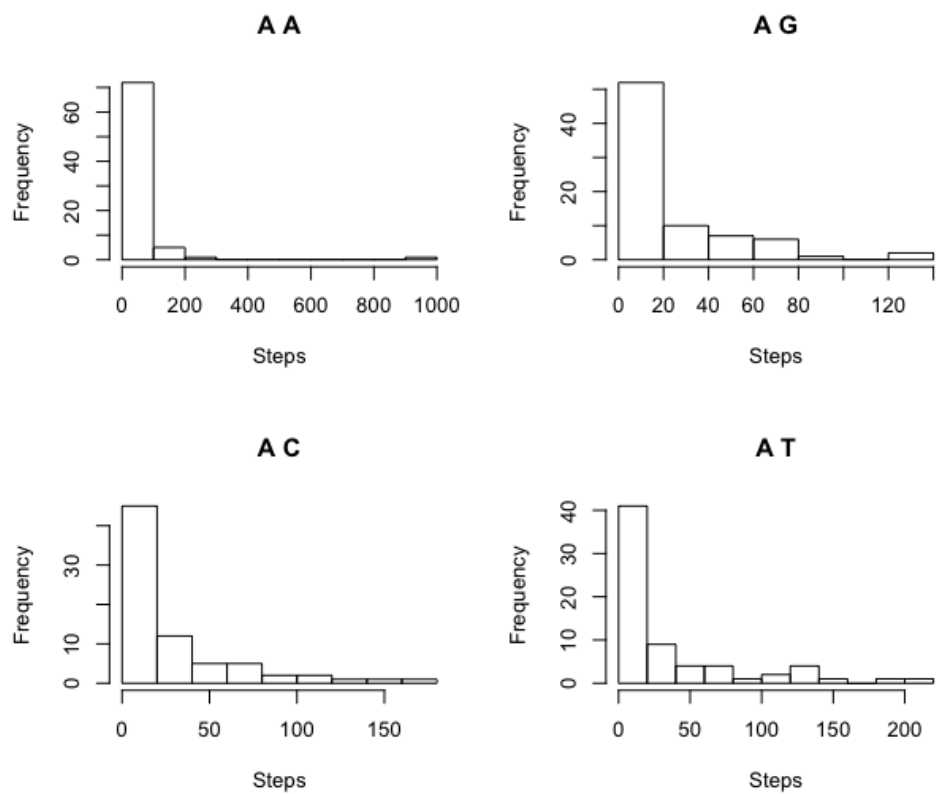


Figure 27

Comments:

The return time distributions of all the pairs of nucleotides (dimers) for all the sequences is observed to be same. There is no significant difference between all the sequences, which is evident from the entropy.

Tree Topology using Return Times:

To obtain Tree topology from Return Time distribution, following distance function is used.

$$d_{ij} = + \sqrt{\sum_{r \in S} (\mu_r^i - \mu_r^j)^2 + (\sigma_r^i - \sigma_r^j)^2}$$

Where,

$S = \{A, G, C, T\}$, μ_r^i : Mean of the distribution of r^{th} element in S of i^{th} sequence

σ_r^i : Std. deviation of the distribution of r^{th} element in S of i^{th} sequence

Means and Standard Deviations for each distribution along with distance matrix are given below.

Mean:

	A	G	C	T
1	2.343292	4.095563	4.437079	2.112802
2	2.348923	4.095051	4.437682	2.111725
3	2.349192	4.093718	4.436692	2.112849
4	2.342547	4.093873	4.437887	2.113591
5	2.343068	4.094742	4.439060	2.112710
6	2.342547	4.093873	4.436897	2.113916
7	2.343292	4.094864	4.437079	2.112802

Std. deviation:

	A	G	C	T
1	2.943416	4.569176	4.875107	2.681158
2	2.944952	4.569466	4.875536	2.680083
3	2.945267	4.566612	4.874598	2.681016
4	2.943042	4.567396	4.875183	2.681495
5	2.943518	4.568049	4.875488	2.677430
6	2.942966	4.566088	4.875590	2.680395
7	2.943416	4.569099	4.875107	2.680614

Distance Matrix:

	1	2	3	4	5	6	7
1	0.000000	0.006106	0.006975	0.002848	0.004470	0.003904	0.000889
2	0.006106	0.000000	0.003754	0.007458	0.006954	0.007932	0.006019
3	0.006975	0.003754	0.000000	0.007232	0.007935	0.007232	0.006805
4	0.002848	0.007458	0.007232	0.000000	0.004522	0.002044	0.002576
5	0.004470	0.006954	0.007935	0.004522	0.000000	0.004485	0.003924
6	0.003904	0.007932	0.007232	0.002044	0.004485	0.000000	0.003516
7	0.000889	0.006019	0.006805	0.002576	0.003924	0.003516	0.000000

Tree topology is obtained using NJ method.

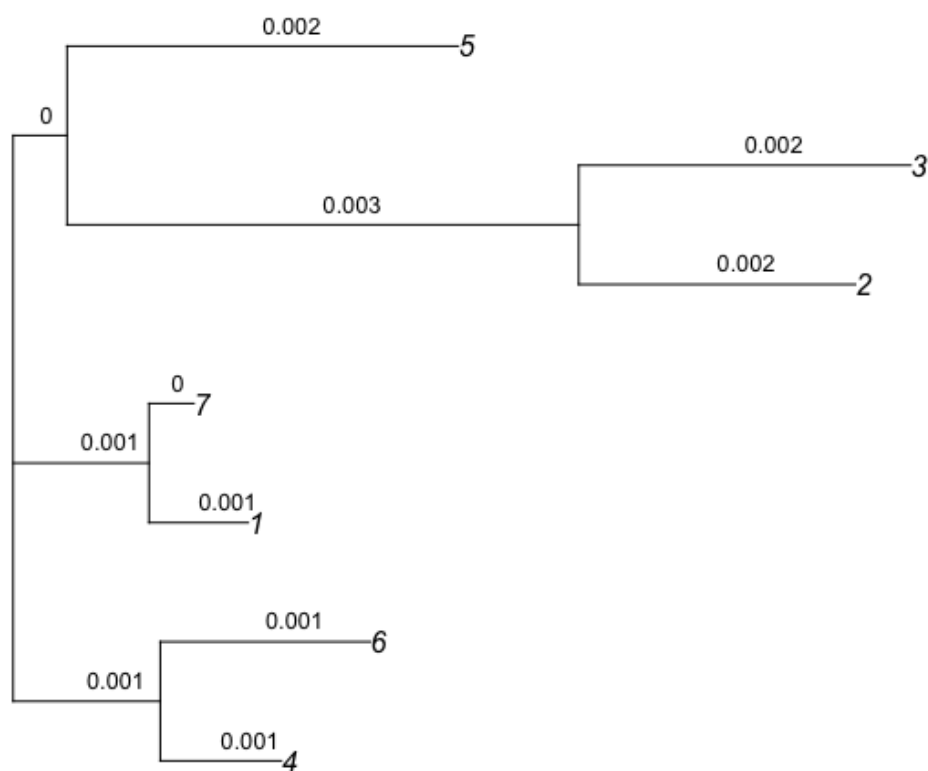


Figure 28

Comments:

The NJ method using the given distance metric gives an unrooted phylogenetic tree. The sequences 1 and 7 are observed to be the most similar here.

Question 6

- Aim:** i) To obtain one step TPMs for each sequence
 ii) To check whether the Markov Chains are ergodic or not.

Estimation of TPM:

Given a sequence of length n , the transition probability matrix (TPM) is estimated as follows :

Step 1 :

Construct a 4×4 matrix of zeroes, with rows and columns corresponding to each nucleotide {A,T,C,G}.

Step 2 :

Consider the nucleotide in 1st place of the sequence, say it is A. If the nucleotide in the 2nd place is G, then increase the value in the place corresponding to row A and column G.

Similarly, if the nucleotide at the i^{th} place ($i = 1, 2 \dots n-1$) is S_1 and that at the $(i+1)^{\text{th}}$ is S_2 , then the value corresponding to the S_1^{th} row and S_2^{th} column is increased by 1. ($S_1, S_2 \in \{A, T, C, G\}$). This step results in the transition count matrix.

Step 3 :

Divide all the rows of the transition count matrix by the respective row totals, to get the transition probabilities.

The resultant matrix is the required TPM.

The TPMs for all the sequences are calculated using the above procedure.

Let $S = \{A, C, G, T\}$ with initial distribution $P(X = i) = 1/4$, $\forall X \in S$

Now, estimate of one step TPM is given by,

$$\hat{P} = \left((\hat{P}_{ij}) \right) \text{ Where, } \hat{P}_{ij} = \frac{n_{ij}}{n_i} = \frac{\text{No of Transitions from 'i' to 'j'}}{\text{Total number of 'i's in sequence}}$$

For all the sequences, TPM estimates are obtained using an R program.

[[1]]

	A	G	T	C
A	0.3206893	0.19480810	0.2263623	0.2581403
G	0.2750384	0.18631633	0.1992834	0.3393619
T	0.3794610	0.07993445	0.1616897	0.3789148
C	0.2476806	0.26978005	0.1472949	0.3352444

[[2]]

	A	G	T	C
A	0.3192778	0.19535718	0.2265336	0.2588314
G	0.2747440	0.18651877	0.1993174	0.3394198
T	0.3793041	0.07997814	0.1615959	0.3791219
C	0.2478116	0.26959150	0.1473531	0.3352439

[[3]]

	A	G	T	C
A	0.3191967	0.19522046	0.2268596	0.2587232
G	0.2746671	0.18641174	0.1993855	0.3395357
T	0.3792350	0.07996357	0.1617486	0.3790528
C	0.2477839	0.26989259	0.1472521	0.3350714

[[4]]

	A	G	T	C
A	0.3207252	0.19482990	0.2262757	0.2581692
G	0.2750853	0.18634812	0.1993174	0.3392491
T	0.3795301	0.07994901	0.1615371	0.3789838
C	0.2477055	0.26992073	0.1473717	0.3350021

[[5]]

	A	G	T	C
A	0.3208012	0.19480810	0.2261385	0.2582522
G	0.2751323	0.18655061	0.1990101	0.3393070
T	0.3794171	0.07959927	0.1617486	0.3792350
C	0.2477064	0.26980817	0.1475188	0.3349666

[[6]]

	A	G	T	C
A	0.3207252	0.19482990	0.2261638	0.2582811
G	0.2750853	0.18634812	0.1991468	0.3394198
T	0.3794610	0.07993445	0.1616897	0.3789148
C	0.2477313	0.26994889	0.1475957	0.3347241

[[7]]

	A	G	T	C
A	0.3206893	0.19480810	0.2263623	0.2581403
G	0.2749915	0.18645513	0.1990788	0.3394746
T	0.3796431	0.07993445	0.1615076	0.3789148
C	0.2475764	0.26978005	0.1475034	0.3351402

Ergodic or not?

Ergodic Markov Chain is a Markov Chain in which all states are persistent non-null and aperiodic.

All our sequences are finite Markov Chains. Also, we can see that, for all the sequences all the entries in one step TPM are positive, which implies aperiodic and persistent non-null nature of all states. Hence, all the sequences are Ergodic Markov Chains.

Bonus Question

Using the estimated TPMs given on page no. 29-30, 3 copies of each sequence are simulated. The simulations are carried out using the following steps :

Step 1 :

Assuming probability of having any particular one from the four nucleotides {A, G, T, C} at the first place is $\frac{1}{4}$, a nucleotide is selected randomly.

Step 2 :

Let the nucleotide selected at the first place be S_1 ($S_1 \in \{A, T, C, G\}$). For the second place, select a sample of one from the set of all four nucleotides, with probability corresponding to the row S_1 in the TPM. Let the nucleotide selected for the second place be S_2 ($S_2 \in \{A, T, C, G\}$).

Step 3 : Repeat step 2 until the sequence of length 1000 is obtained.

The resultant sequence is the required simulation of the biological sequence, given the TPM. 3 such copies for each TPM are computed.

TPMs for these simulated sequences computed using the steps mentioned in Q6. As an example, first 100 terms of copies of 1st sequence are given below, each of size 1000.

Copy 1:

"A" "T" "A" "T" "T" "T" "C" "T" "A" "A" "A" "A" "C" "G" "A" "A" "A" "C" "C" "G" "T"
 "A" "C" "A" "A" "C" "T" "A" "T" "A" "T" "A" "A" "A" "T" "T" "A" "T" "A" "A" "T" "A" "T"
 "C" "T" "T" "A" "G" "T" "A" "A" "T" "C" "C" "A" "G" "A" "A" "A" "G" "A" "C" "A" "C"
 "C" "G" "C" "A" "C" "C" "A" "A" "T" "A" "T" "C" "T" "C" "T" "A" "C" "T" "T" "C" "G" "A"
 "T" "G" "T" "T" "T" "T" "T" "C" "T" "G" "T" "A" "T" "T" "A" "G" "T" "G" "T" "A" "G" "T"
 "A" "A" "A" "T" "T" "C" "T" "A" "C" "T" "A" "A"

Copy 2:

"A" "A" "T" "T" "A" "G" "A" "T" "C" "G" "A" "A" "T" "C" "A" "A" "T" "A" "G" "C" "T"
 "A" "G" "A" "C" "T" "A" "C" "C" "G" "C" "C" "G" "G" "T" "C" "T" "T" "A" "T" "A" "T" "A"
 "G" "C" "A" "G" "A" "C" "G" "T" "T" "A" "T" "T" "A" "A" "G" "T" "A" "T" "G" "A" "C" "T"
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 "T" "A" "A" "G" "G" "T" "T" "A" "C" "T" "C" "T" "A" "C" "G" "C" "A" "T" "C" "T" "T" "T"
 "A" "A" "C" "C" "A" "A" "G" "C" "A" "T" "G" "A"

Copy 3:

"G" "G" "T" "A" "A" "A" "C" "C" "A" "T" "C" "T" "T" "A" "T" "A" "C" "A" "T" "C" "T" "A"
 "A" "G" "T" "A" "G" "C" "G" "A" "T" "T" "A" "A" "C" "T" "G" "A" "C" "G" "C" "A" "A"
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 "T" "G" "A" "G" "C" "T" "T" "G" "G" "A" "A" "T" "A" "C" "A" "T" "T" "T" "T" "C" "T" "T"
 "T" "C" "C" "A" "C" "G" "A" "A" "C" "A" "A"

Re-estimation of TPMs**Copy 1:**

	A	G	T	C
A	0.3374233	0.19631902	0.2208589	0.2453988
G	0.3202247	0.15168539	0.1910112	0.3370787
T	0.2584270	0.32584270	0.1123596	0.3033708
C	0.3596215	0.08832808	0.1640379	0.3880126

Copy 2:

	A	G	T	C
A	0.3531157	0.1869436	0.2106825	0.2492582
G	0.2317073	0.1768293	0.2195122	0.3719512
T	0.3483146	0.2584270	0.1292135	0.2640449
C	0.3687500	0.0843750	0.1500000	0.3968750

Copy 3:

	A	G	T	C
A	0.2491582	0.19191919	0.2693603	0.2895623
G	0.3048780	0.17682927	0.1829268	0.3353659
T	0.2383420	0.24352332	0.1139896	0.4041451
C	0.3681159	0.08695652	0.1768116	0.3681159

Comments:

The estimated TPMs of the simulated sequences are observed to be close to the original sequence.

Concluding Remarks

- 1) All the sequences contain almost same level of information. Furthermore, mutual information between each pair of sequences is also same.
- 2) In most of the tree topologies, Sequences 1 and 7 are clustered together. Few topologies have shown equal level of similarities between the sequences with almost equal branch lengths.
- 3) All the distance functions which are chosen for NJ method do not satisfy ultrametric condition.
- 4) Return time distributions of mono and di levels are almost same for all the sequences. For mono level, the distribution is turned out to be geometric.
- 5) If we consider each sequence as a Markov Chain, then we have shown that all of those are Ergodic i.e. nicely behaved Markov Chains.

Computational Time:

Following is the time required for the computations made for this assignment on macOS Sierra, 1.6 GHz Intel Core i5 with 8 GB 1600 MHz DDR3 memory.

No.	Question	Task	Time (in secs)
1	-	Data import and formatting	0.046767
2	1	Computation Entropy	1.154186
3	1	Computation of MIC	16.06423
4	2	UPGMA using Entropy based distance	19.90745
5	3	Tree Topology using MIC based distance	0.02894306
6	4	NJ Method	0.5158439
7	5	RTD for mono level with Tree	1.068517
8	5	RTD for di level	1.683077
9	6	TPM Estimation	2.511241
10	Bonus	Markov Chain Simulation and Re-estimation	0.5195401

Total time required for running the code of entire assignment is **1.247577 mins.**

References

- 1) www.evolution-textbook.org
- 2) GISAID - <https://www.gisaid.org/>
- 3) NCBI - <https://www.ncbi.nlm.nih.gov/>
- 4) Pandurang Kolekar, Mohan Kale and Urmila Kulkarni-Kale, “Molecular Evolution & Phylogeny: What, When, Why & How?”, Computational Biology and Applied Bioinformatics, Edited by Heitor Lopes, 2012, DOI: 10.5772/772
- 5) N Saitou, “The neighbour-joining method: a new method for reconstructing phylogenetic trees”, *Molecular Biology and Evolution*, Volume 4, Issue 4, Jul 1987, Pages 406–425.